

# Research Assignment 2 Module 1

## The Language of Business Analytics

### Question 1

#### 1) Business Analytics: The discipline itself

Most modern organizations are completely swimming in metrics yet starving for actual direction. This exact paralysis is what business analytics tackles head-on. Rather than wasting hours staring aimlessly at massive spreadsheets, analytics teams deploy targeted quantitative models to pinpoint past operational failures and map out next quarter's revenue trends. Think of it as a rare hybrid of software engineering and raw commercial strategy that answers two fundamental questions like: What happened, and what do we do next?

To distinguish Business Analytics from Data Analytics and Business Intelligence, Business Analytics decides where to steer the car next or what to do next.

#### 2) Data Analytics: Often confused with Business Analytics

At its core, data analytics is just about interrogating raw information until it tells you something useful. Today, most of the heavy lifting like scrubbing messy files and parsing datasets happens through automated scripts and algorithms that chew through data so that humans don't have to. Ultimately, the goal is simple: pull hidden operational trends out of the noise and turn them into concrete tweaks that make a business run faster, cheaper and smarter. Data Analytics can be distinguished from Business Intelligence and Business Analytics by the



fact that it builds the engine.

3) Business Intelligence (BI): Reporting & insight tools: Think of business intelligence as a company's operation mirror. It is the practice of gathering scattered internal metrics and turning them into clear, visual layouts so management can actually see how the business is running day-to-day, rather than guessing or assuming. While business analytics focuses heavily on forecasting future trends, BI is strictly about nailing down the present moment. It provides organizations the exact clarity they need to streamline messy workflows, spot hidden market opportunities, and drive consistent operational growth. To further distinguish it from Business Analytics and Data analytics, BI looks at the dashboard to see how fast you are going.

#### 4) Data-Driven Decision Making: A modern management philosophy

At its core, data-driven decision-making means steering an organization using hard evidence rather than raw intuition or gut feeling. Instead of gambling on informal judgement, leaders let systematic data analysis dictate their strategic choices. In the real world, this looks like tracking market shifts, breaking down customer feedback loops, and auditing financial metrics to see what is actually happening under the hood.

For example, instead of relying on a gut feeling to stock shelves, Shoprite used hard data from Xtra Savings Card swipes to automate their supply chain strategy.



## 5) Big Data: Volume, Velocity, Variety:

Big Data is data that is simply too fast, too messy, and too massive to handle using old-school tech tools. It is an overwhelming wave of messy information that is simply too fast and massive for normal spreadsheets to handle. Once cleaned, this chaotic data exposes hidden blind spots, turning an IT nightmare into a goldmine for making smart corporate choices. For example: Discovery Vitality tracks millions of live gym swipes, Apple Watch heart rates, and driving telematics every second. Organizing this chaotic data wave exposes hidden health and accident trends, allowing them to adjust insurance premiums and offer real-time rewards that nudge members into safer habits.

## 6) Descriptive Analytics: "What happened?"

Descriptive Analytics looks backward to show exactly what already happened inside a business, by tracking historical data, like comparing quarterly sales or year-over-year revenue dips. It packages past metrics into clear facts so leaders know exactly where the company stands today. For example, Woolworths reviews past register data to compare seasonal sales against previous years. This ignores forecasting entirely, giving executives a clear look at which branches hit their targets today.

## 7) Diagnostic Analytics: "Why did it happen?"

Diagnostic Analytics digs into sudden changes in company numbers to uncover the exact why. It runs data drill-downs and correlation checks to move past guesswork, isolating the specific root cause like a broken website checkout page or a shipping delay.



For example, Takealot analyzes sudden drops in sales by examining server logs and payment gateways to isolate technical glitches. This diagnostic approach allows engineers to immediately resolve software bugs and checkout timeouts during peak shopping hours.

### 1) Predictive Analytics: "What is likely to happen?"

Predictive Analytics acts as a company's radar by searching historical records to forecast future trends. By running machine learning and statistical models on existing customer data, it converts dormant information into forward-looking insights that help executives make proactive strategic moves.

### What Predictive Analytics is not:

a) Not a crystal ball: It calculates statistical probabilities based on past data, not absolute guarantees; unforeseen disruptions or challenges that cause the models to fail.

b) Not descriptive analytics: It does not report past facts; descriptive analytics tells you who defaulted last month, while predictive estimates who might default next month.

c) Not prescriptive analytics: It forecasts what will likely happen but does not provide the actual solution or automate the fix.

d) Not raw intuition: It relies strictly on clean, structured data pipelines, completely failing if fed biased information or gut guesswork.

An Example: FNB uses machine learning to analyze historical spending and debit order patterns, catching



early indicators of financial distress to predict loan defaults before they occur.

### a) Prescriptive Analytics: "What should we do about it?"

Prescriptive analytics goes beyond spotting patterns by recommending the exact, step-by-step actions to take. By using advanced algorithms to weigh risks, it provides a clear game plan for any scenario, allowing companies to automate decisions and handle disruptions before they happen.

### What Prescriptive Analytics is not:

a) Not just a prediction: It doesn't stop at forecasting what will happen; it goes a step further by actively commanding the best route or operational change to make.

b) Not a boardroom brainstorm: It does not rely on executive debates or human intuition; choices are driven entirely by data-fed optimization engines.

c) Not a static report: It isn't a fixed monthly dashboard; it processes live inputs in real time to update action plans dynamically.

d) Not foolproof: It is completely dependent on accurate rules; if the underlying data or code assumptions are flawed, it confidently automates terrible operational decisions.

An Example: Checkers Sixty60 automates driver dispatching and routing by processing live data on vehicle locations, traffic and load shedding to push real-time turn-by-turn routes to drivers' phones, guaranteeing 60 minutes delivery windows while cutting fuel costs.



## Question 2

### 11) Structured Data: Rows and Columns

Structured Data organizes information into clean rows and columns. Tracking this data generates metrics, any measurable number, like daily websites hits. However, every KPI is a metric, but not every metric is a KPI. While a metric is just a plain spreadsheet number, a KPI is a hyper-specific metric that directly tracks whether a business is succeeding or failing at its core strategic goal.

#### An example:

a) Pick a Pay (Metric): Tracks electricity kilowatt-hours for routine utility bill payments. It is just an operational cost; a spike doesn't measure strategic growth.

b) Solaroff (KPI): Tracks these exact same kilowatt-hours as a high-stakes survival metric. For a solar energy provider, output efficiency directly dictates revenue and contract compliance.

### 12) Unstructured Data: Text, images, audio, video

Unstructured data is messy information like text, audio, and video that won't fit into neat database rows. Extracting clean numbers from this chaos creates metrics. While a metric is just a baseline data point (like total audio file length), a KPI is a high-stakes number that directly tracks whether a business is hitting its core strategic goals.

An Example: a) Bowmans Law (Metric): Tracks audio minutes for server storage; longer files do not boost legal profits.

b) CI (Call Centre KPI): Tracks these exact same audio minutes because talk time directly dictates billing and



Client contract compliance.

### 13) Data Warehouse: Central Storage for Analytics

Data warehouses are centralized storage hubs built to consolidate messy operational data from different departments into a single source for analytics. Pulling data points from this hub creates metrics. While a metric is just a baseline data point in cold storage (like total storage bytes utilized), a KPI is a high-stakes number that directly tracks whether a business is hitting its core strategic goals.

An Example: a) Pickn Pay (Metric): Tracks database queries to monitor hardware performance. It is an operational cost; more queries do not boost super-market profit.

b) Sizwe IT (KPI): Tracks these exact same queries as a high-stakes metric. Computational query counts directly dictate customer billing milestones and corporate revenue.

### 14) Data Mining: Discovering patterns

Data mining is the computational process of digging through massive datasets to uncover hidden patterns and correlations. Extracting these trends creates metrics. While a metric is just a baseline data point discovered during mining (like database scanning speed), a KPI is a high-stakes number that directly tracks whether a business is hitting its core strategic goals.

An Example: a) Juppiechef (Metric): Mines sales correlations for web layouts; finding more product pairs does not drive retail growth.



b) Time Bank (KPI): Nines deposit-to-repayment correlations; this behavioural score is a high-stakes metric that automates real-time credit approvals

### 15) Data Visualization: Charts, graphs, infographics

Data visualization converts complex datasets into visual assets like charts and graphs to spot trends at a glance. Translating numbers into these visual elements creates metrics. While a metric is just a baseline data point mapped onto a chart (like server rendering time), a KPI is a high-stakes number that directly tracks whether a business is hitting its core strategic goals.

### 16) Dashboard: A live <sup>view</sup> of performance

A dashboard is a live console displaying real-time operational data. Linking data pipelines to these automated graphs exposes performance patterns, known as metrics. A basic metric is simply a minor background number tucked away on a sub-tab (like hardware ping speeds), a KPI gets prime placement on the main screen because it directly proves whether a company is reaching its primary strategic targets.

### 14) Metric: Any measurable number

A metric is any quantifiable data point or measurable number tracked to monitor daily workflows, like office water usage or gym swipes. It captures basic operational facts without adding strategic context.

A standard metric is just a routine number on a spreadsheet, a KPI is a high-stakes, curated data point that directly tracks where a business is succeeding or failing at its core strategic goals.



## 18) Key Performance Indicator (KPI): A metric that drives decisions

It is a critical metric that tracks whether a business is achieving its core goals, serving as a primary decision driver for immediate corporate pivots. A baseline metric is just any routine spreadsheet number (like emails sent), a KPI is a curated selection that directly dictates business survival and revenue execution.

## 19) Benchmark: A reference point for comparison

A Benchmark is an external industry standard used to check how your company's data measures up against competitors. A basic baseline metric is really just a standard day-to-day number you keep an eye on like monitoring your server uptime. On the other hand, a KPI is a high-stakes, heavily benchmarked figure that tells you exactly whether the business is actually hitting its biggest strategic milestones.

## 20) Segmentation: Grouping customers by attributes

Segmentation breaks a messy customer base down into smaller, distinct pockets based on common threads like geographic location or spending habits. When you start tracking data within those specific groups you get your metrics. It is also worth noting that while every single KPI is a metric, the reverse is not true. A baseline metric is really just a standard day-to-day number you pull from a specific customer pocket such as counting how many high-income users you have. In contrast, a KPI is a high-stakes, highly segmented figure that tells you exactly whether the business is reaching its biggest strategic goals.



### Question 3

#### 21) Conversion Rate: % of visitors who take action

Conversion rate provides a business with a percentage of visitors who performed a specific desired action such as placing an order, subscribed or making a purchase. Due to the fact that it is not simple to tell if a business is going to succeed without comparing the gaps between what a business pays to win customers (CAC) and what the customer spends in a business (CLV).

This comparison provides a business with a clear moment of truth.

a) When a business looks at the Customer Lifetime Value (CLV) to Customer Acquisition Cost (CAC) ratio for comparison. It is in a great spot if the ratio is 3:1 or up. It shows that the business making profits because customers are bringing in more money than what a business spent to win those customers.

b) If the ratio is 1:1 or lower it means that a business is using and losing more money on marketing campaigns that are not bringing in customers in order to make a profit. This information can help a business to come up with strategic plan.

For Example: For an online clothing store or brand, a business can check its website analytics in order to see if profits are coming in for that particular month and the following data is what would appear:

- Total Number of Website Visitors (Traffic): 40,000 people visited the website



- Total Conversions (Finalized Actions): 600 people made an actual purchase.

## 22) Customer Acquisition Cost (CAC): Cost of gaining one new customer.

Customer Acquisition Cost (CAC) measure how much a business spent (particularly in marketing campaigns) in order to gain or acquire a new customer.

For Example: Amount spent in total = R40,000.  
Gained/New Customers = 80  
Calculation =  $\frac{R40,000.00}{80} = 500$

It means that CAC is R500 per customer.

## 23) Customer Lifetime Value (CLV/LTV)

Customer Lifetime Value (CLV) is a total amount of money or profit made from one customer while they do business with you (as an organization or company). For example, let's say you own a baked goods store with delectable scones, muffins, cakes, etc, and you have one loyal customer who visits your store 4 times a week to spend R50 daily at your store. This customer is roughly at your store 200 times a year and they remain to be a loyal customer for 4 years until they relocate to a different country or city, it means the CLV gained from that client is R40,000 ( $80 \times R500 \text{ visits} \times 4 \text{ years} = R40,000.00$ )

## 24) Churn Rate: % of customers who leave

Churn rate is a percentage of customer who stop using services of your business or stop buying from your business within a specific period or timeframe.



## 25) Customer Retention Rate: The opposite of churn

Customer Retention Rate (CRR) is the percentage of who keeps buying from the business or who already are existing and keep purchasing from the business. For example: Imagine a modeling agency membership with 100 members at the beginning of the month, 105 members at the end of the month and 15 new members.

Total members increased by 105 and the agency lost 10 of its loyal or original members which resulted in a 90% retention rate and a 10% churn rate. CRR is the opposite of churn rate in this sense.

## 26) Net Promoter Score (NPS): Would your customers recommend you?

Net Promoter Score (NPS) helps a business track loyal and how happy and satisfied customers are with the services and products by asking them an important and direct question such as "How likely are you to recommend our business services or products to others?" This survey will help a business fix their problems to ensure an overall satisfaction from most customers.

## 27) Average Order Value (AOV): Revenue per transaction

Average Order Value (AOV) tracks or measures the amount of money a customer spent on a single transaction or purchase online or in store.

## 28) Sales Funnel: From awareness to purchase

It's a process a buyer takes from discovering your brand to buying it, or complete a purchase. For example, a buyer starts browsing your business online and ends up buying your services or products.



## Question 4: Marketing and Campaigns Analysis

### 29) Marketing Campaign: A coordinated marketing effort

Marketing Campaign is a core initiative used to promote a business's goal or objective, by effectively communicating with the target audience in order to increase brand awareness, improve visibility and gain new clients.

Interconnection: It creates the baseline data that the Post-Campaign Analysis will later measure. For example, a campaign launch of Black Friday promotion email, text and social media platform that makes clients want to click on, view, sign-up for and finally make a purchase, will later be used by the Post-Campaign Analysis to check if a goal was achieved.

### 30) Post-Campaign Analysis: Meaning what worked

It is not just about reviewing numbers, it helps businesses make smarter decisions in the future using data obtained from an overview of what worked, what didn't work and why it didn't work in a marketing campaign that was held. To simplify this definition, it is a data review done after a marketing campaign project ends in order to evaluate how well the core initiative was executed.

Interconnection: It uses metrics to measure financial success. Such metrics may be Incremental Revenue ROI and ROAS used to calculate the financial impact effectively.



### 31) Incremental Revenue: Extra revenue caused by the campaign

This is an extra profit a business made from a certain marketing campaign or promotional campaign.

Interconnection: It serves as the initial metric in a Post-Campaign Analysis to help measure the Return On Investment (ROI).

### 32) Return on Investment (ROI): The Classic Profitability Test

It measures how much profit a business has made compared to how much a business spent.

Interconnection: It compares a business extra profit or sales (Incremental Revenue) against the total cost on a campaign, spent by a business.

This number is the final indicator of whether the campaign project made money or not and that is evaluated in a Post-Campaign Analysis.

### 33) Return on Ad Spend (ROAS): Revenue per advertising dollar

It is a financial and marketing performance metric that evaluates the efficiency of advertising campaigns by measuring how much gross revenue a business earns for every amount spent on advertising (Bailey 2025).

Interconnection: ROI evaluates macro-level profitability, whereas ROAS tracks immediate media performance. For example, if A/B testing reveals that a creative concept yields a higher click-through rate, that specific variation generally produces a superior ROAS.



### 34) Click-Through Rate (CTR): % of viewers who click

Click-Through Rate (CTR) is a digital marketing performance metric defined as the mathematical ratio of total clicks to total impressions generated by a specific online asset, such as an email, search result, or mobile advertising campaign (Insider One 2024).

Interconnections: It shows and measures early creative engagement, but its value depends on keeping the Bounce Rate low and driving successful Lead Generation.

### 35) Bounce Rate: % of visitors who leave quickly

It is the percentage of visitors who browse through a specific business page and exit immediately without clicking anything else or browsing any further.

Interconnection: A high Bounce Rate alongside a high CTR shows a gap between ad expectations and the page content. Businesses therefore use A/B Testing to fix these friction points.

### 36) A/B Testing: Comparing two versions

According to Fullstory (2024), A/B Testing - interchangeably referred to as split testing or bucket testing, is defined as a digital optimization methodology that involves presenting two distinct iterations of an asset simultaneously to comparable groups of web visitors to determine which variation yields a superior performance metric.

Interconnection: It provides the insights that are needed to lower bounce rates, boost CTR, and increase lead generation.



### 37) Lead Generation: Capturing potential customers

Mohandas (2025) stated that Lead Generation is the operational process of stimulating consumer interest, attracting web visitors, and capturing inquiries regarding a company's products or services to convert them into potential customers.

Interconnection: The count of new prospects serves as a post-campaign performance benchmark, acting as the pipeline that drives incremental revenue and a healthy ROI.

### 38) Marketing Attribution: Which channel deserves the credit?

As outlined by Brauning (2025), marketing attribution is a data analysis framework used to identify which specific marketing efforts drive business results by systematically assigning fractional credit or value to distinct customer interactions known as touchpoints along the path to a conversion.

Interconnection: It provides granular data for post-campaign analysis, breaking down incremental revenue and ROAS by channel to guide future budget allocations.